

SFT V11.5: The One-Speed Universe

The Causal Budget of Space, Time, and Expansion

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Abstract

The speed of light is not fundamentally a speed. It is the universal causal processing rate — the ratio of one Planck length to one Planck time, one spatial operation per temporal operation, one tick per tick. That it manifests as the measured speed of photons is a consequence, not the definition. Every persistent physical process in the universe runs at this rate. A photon expresses it as spatial propagation. A massive particle expresses it through a combination of spatial translation and internal phase evolution. A particle at rest expresses it as pure proper-time advance and internal Compton cycling. Special relativity, quantum mechanics, gravity, and cosmic expansion are different accountings of how this single universal rate is distributed across spatial, internal, and temporal directions. This note states the one-speed principle, shows that it is equivalent to the Compton Maintenance Postulate, and identifies the sense in which c is not a law of physics but an identity: one operation takes one operation.

1 One Speed

Define the Planck length and Planck time:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}. \quad (1)$$

Their ratio is

$$\frac{\ell_P}{t_P} = c. \quad (2)$$

This ratio does not derive c from deeper constants — the Planck units are constructed from c in the first place. What it shows is that in the natural unit system where quantum, gravitational, and causal scales meet, one spatial Planck unit per one temporal Planck unit is exactly c . SFT interprets that identity as the elementary causal processing rate:

$$c = \frac{\text{one spatial operation}}{\text{one temporal operation}} = \frac{\text{one pixel}}{\text{one tick}} = 1 \frac{\text{operation}}{\text{operation}} \quad (3)$$

The Planck length is not a small distance that happens to have a special value. It is the spatial extent of one elementary operation. The Planck time is not a small duration that happens to equal ℓ_P/c . It is the temporal extent of the same operation. Their ratio is c by construction — but SFT reads that construction as physically meaningful: c is the rate at which the elementary operation executes, and the Planck units are its spatial and temporal footprints.

2 Everything Moves at c

If c is the universal processing rate — one operation per tick — then every physical process runs at this rate. The question is not how fast something moves. The question is where it allocates its operations.

2.1 A photon

A photon allocates all of its operations spatially. Each tick advances it one Planck length through space. It has no internal cycling. It has no rest mass. Its entire c -budget is spent on spatial propagation.

$$v_{\text{spatial}} = c, \quad v_{\text{internal}} = 0, \quad m = 0. \quad (4)$$

2.2 A massive particle at rest

A massive particle at rest allocates all of its operations internally. Each tick advances its phase by one increment of the Compton cycle. It does not move through space. Its entire c -budget is spent on internal phase evolution.

$$v_{\text{spatial}} = 0, \quad v_{\text{internal}} = c, \quad \nu_C = \frac{mc^2}{h}. \quad (5)$$

The internal cycling at rate ν_C IS the mass. The particle is “at rest” in space but moving at c through its internal phase. It is writing to the same local patch of the holographic boundary, tick after tick, at the full processing rate.

2.3 A massive particle in motion

A massive particle moving at velocity v splits its causal budget between spatial advance and internal phase evolution. The budget partition is expressed cleanly in dimensionless form. Define the spatial fraction and the internal/proper-time fraction:

$$\beta_{\text{space}} = \frac{v}{c}, \quad \beta_{\text{phase}} = \frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}. \quad (6)$$

These satisfy the causal budget identity:

$$\boxed{\beta_{\text{space}}^2 + \beta_{\text{phase}}^2 = 1} \quad (7)$$

This is Pythagoras applied to the causal budget. The spatial fraction and the proper-time fraction are two legs of a right triangle whose hypotenuse is unity — the full budget. Note that β_{phase} is not a second ordinary velocity through hidden space; it is the fraction of the causal budget remaining for proper-time advance and internal phase evolution.

More spatial motion means less proper-time advance per unit coordinate time. Less proper-time advance means fewer Compton cycles registered relative to the external frame. The rest mass is unchanged; the coordinate-observed ticking rate is time-dilated:

$$\nu_{\text{obs}} = \nu_C \frac{d\tau}{dt} = \frac{mc^2}{h} \sqrt{1 - \frac{v^2}{c^2}}. \quad (8)$$

This is time dilation, understood not as a mysterious relativistic effect but as causal-budget reallocation: motion through space redirects part of the processing budget from internal phase advance to spatial propagation.

2.4 The invariance of proper velocity

In special relativity, the four-velocity of any massive particle satisfies

$$u^\mu u_\mu = -c^2. \quad (9)$$

This equation is usually presented as a mathematical constraint on the four-velocity vector. In SFT, it has a physical meaning: the total processing rate is c , always, for everything. The four-velocity constraint is the conservation of processing speed. It cannot be violated because there is nothing to violate — c is the only rate that exists.

3 Why Nothing Exceeds c

In the standard presentation, the speed of light is a speed limit. Nothing with mass can reach it. Nothing without mass can avoid it. The limit is postulated as a fundamental law.

In SFT, the speed of light is not a limit. It is the only speed there is.

A particle cannot exceed c for the same reason that a fraction cannot exceed 1. If the particle is already allocating all of its operations — its entire c -budget — then there are no additional operations to allocate. “Faster than light” would mean processing more than one operation per tick. But a tick IS one operation. You cannot fit two operations into one operation. It is not a physical constraint. It is a logical impossibility.

This is why accelerating a massive particle toward c requires infinite energy. As $v_{\text{spatial}} \rightarrow c$, the internal cycling rate $v_{\text{internal}} \rightarrow 0$. The particle’s internal phase structure is being squeezed toward zero. To reach c exactly, the internal cycling must stop entirely, which means the particle must become massless — a complete restructuring of its internal phase architecture. That costs infinite energy because you are trying to convert a finite-mass cycling process into a zero-mass spatial mode.

4 Time Dilation as Budget Reallocation

A clock on a spaceship moving at velocity v runs slow by the Lorentz factor:

$$\Delta t' = \gamma^{-1} \Delta t = \Delta t \sqrt{1 - v^2/c^2}. \quad (10)$$

This is universally confirmed. The standard explanation is that time itself flows differently in different frames. The SFT explanation is simpler:

The clock is a ticking process. Moving the clock spatially redirects part of its c -budget from internal ticking to spatial propagation. The clock ticks slower because it is spending some of its operations on moving instead of ticking.

The clock is not affected by some mysterious relativistic field. The clock is a particle — or a collection of particles — ticking at the Compton rate. Move it through space and some of that Compton ticking converts into spatial motion. The total stays c . The internal component decreases. The clock runs slow.

This is not a reinterpretation of special relativity. It is special relativity given a physical mechanism. The Lorentz factor $\gamma^{-1} = \sqrt{1 - v^2/c^2} = \beta_{\text{phase}}$ is the fraction of the causal budget remaining for internal processing after the spatial allocation.

5 Gravity as Budget Curvature

Near a massive object, clocks run slow. This is gravitational time dilation, confirmed by GPS satellites and Pound-Rebka experiments. In general relativity, the explanation is that spacetime is curved and the curvature affects the flow of time.

In SFT, the explanation is the same budget logic applied to geometry:

Near a massive object, the local lag field is deeper. The particle must spend more of its c-budget navigating the lag gradient. Less budget remains for internal ticking. The clock runs slow.

Gravitational time dilation and velocity time dilation have the same origin: c -budget reallocation. In one case, the reallocation is caused by spatial motion. In the other, it is caused by the lag geometry. In both cases, the total processing rate remains c . The internal ticking rate decreases because the particle is spending operations on something else — spatial motion or geometric navigation.

The Schwarzschild metric encodes this directly. At radial coordinate r from a mass M , a clock held at fixed r (a static clock) has proper-time rate:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{2GM}{c^2 r}}. \quad (11)$$

The proper time $d\tau$ ticks slower than the coordinate time dt by exactly the factor that measures how much of the causal budget is consumed by the gravitational geometry. In Schwarzschild coordinates, $d\tau/dt \rightarrow 0$ at the horizon $r = 2GM/c^2$ for a static clock held at fixed r . This does not mean local physics stops for an infalling observer — an infalling particle's proper time continues normally through the horizon. It means that the static-coordinate budget description degenerates at the horizon: a clock held stationary at that radius would require its entire causal budget just to maintain its position, with nothing remaining for internal phase advance.

6 Expansion as Total Budget

Every particle in the universe ticks. Every tick writes one Planck area to the boundary. The total number of ticks per unit time across the entire observable universe is the total processing rate. The boundary grows to accommodate it. The growth rate is the Hubble parameter:

$$H = \frac{\dot{S}}{I}. \quad (12)$$

Expansion is not a separate phenomenon from the c -budget. Expansion is the global sum of all c -budget allocations. Every particle spending its c -budget internally contributes to the total maintenance demand. The boundary accommodates that demand by growing. The growth rate is H . The growth happens at the only rate anything can happen: c .

The Hubble flow itself moves at c at the Hubble radius:

$$v_H = H \cdot R_H = H \cdot \frac{c}{H} = c. \quad (13)$$

The boundary of the observable universe recedes at c . Not because it is “moving away” at c , but because the ledger is writing at c — the only rate it can write.

7 The Unified Budget

Everything is c . The differences between phenomena are differences in allocation:

Phenomenon	Budget allocation	Standard name
Photon propagation	All spatial	Speed of light
Massive rest	All internal	Compton ticking / mass
Motion	Split spatial + internal	Velocity / time dilation
Gravitational time dilation	Internal reduced by geometry	Curved spacetime
Expansion	Total internal $\rightarrow$ boundary growth	Hubble rate
Pair production	Spatial $\rightarrow$ two internal channels	$E \geq 2mc^2$
Annihilation	Two internal $\rightarrow$ spatial	$E = 2mc^2$ released

One column. One rate. One universe.

8 What This Note Establishes

The one-speed principle is not a new postulate added to SFT. It is a restatement of what the framework already implies, made explicit:

1. SFT interprets the speed of light as the universal causal processing rate. The Planck-unit ratio $\ell_P/t_P = c$ does not derive c ; it identifies c as the elementary rate at which quantum, gravitational, and causal scales meet.
2. Every persistent physical process expresses this rate. Photons express it as spatial propagation; massive particles express it as proper-time advance and internal phase evolution; motion and gravity alter how the rate is projected into coordinate space, internal phase, and boundary accounting.
3. Mass corresponds to internal causal-budget allocation. The Compton frequency $\nu_C = mc^2/h$ is the rate of internal phase processing.
4. Time dilation — both velocity and gravitational — is causal-budget reallocation. The budget identity $\beta_{\text{space}}^2 + \beta_{\text{phase}}^2 = 1$ quantifies the tradeoff.
5. Nothing exceeds c because the causal budget is normalized to unity. Exceeding it would require more than one operation per operation — a logical impossibility, not merely a physical constraint.
6. Expansion is the global accounting of the total causal budget across all particles, accommodated by boundary growth at the processing rate.

The deepest statement is the simplest:

$$c = 1 \frac{\text{operation}}{\text{operation}} \tag{14}$$

One tick takes one tick. Everything else is geometry.

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